

SSD over an HDD. Among SSDs, PCIe, or Peripheral Component Interconnect Express, has been the gold standard for connecting high-speed components to computer motherboards for over a decade. The introduction of PCIe 4.0 in 2017 marked a significant leap in this technology.

At a glance, both M.2 PCIe 3 and 4 drives appear identical. However, the distinction lies in their connection standards to the motherboard, influencing their data transfer speeds. Gen 4 drives, on paper, boast speeds nearly double that of Gen 3 drives, offering a significant boost, especially during tasks that heavily rely on read and write speeds.

Yet, the real-world implications of this speed difference might not be as pronounced for the average user. Everyday tasks like web browsing, gaming, and document management might not showcase the prowess of Gen 4 drives. And most laptops in the Rs 60,000 price come equipped with a PCIe 3 SSD. Lastly, you just need to make sure that your budget gaming laptop has at least 512GB SSD space or more.

What about cooling?

When assessing the cooling performance of a gaming laptop under Rs 60,000, while various synthetic benchmarks exist, a gaming test is the most relevant. The most straightforward method involves checking the sustained FPS during prolonged gaming sessions. This can be done by playing any game on the laptop for more than 25 minutes and observing any FPS drops. Another approach is to execute gaming benchmarks repeatedly and

check if the average FPS remains consistent after each run. For example, based on my observations, if the FPS decreases by over 5 when running the "Shadow of the Tomb Raider" benchmark more than three times, it indicates the laptop's cooling system might be inadequate.

Stress tests like FurMark can also be beneficial in evaluating your laptop's cooling efficiency. Running FurMark pushes your GPU to its peak performance, producing substantial heat. By tracking your GPU's temperature during this test, you can determine the actual effectiveness of the laptop's cooling mechanism.

If the GPU's temperature escalates excessively without stabilizing, or if the laptop powers off due to overheating, it's a sign that the cooling system might not be performing at its best. On the other hand, if the temperature stays within a safe range and stabilizes, it indicates the cooling system is working efficiently.

Nevertheless, it's crucial to be cautious when utilizing stress tests like FurMark. Extended or continuous testing can overburden the hardware. Hence, it's advisable to vigilantly monitor temperatures and halt the test if they approach dangerous levels.

Upgradeability Check

For those considering budget gaming laptops in the Rs 60,000 price range, many of these devices offer the possibility of upgrading either RAM or SSD. If you're curious about whether your laptop has an available SSD slot without physically inspecting it, the user manual that accompanies your laptop

might provide this detail. If the manual is unavailable, an online search for your laptop's specifications might yield results, as most manufacturers list such details on their official websites. If online searches prove fruitless, reaching out directly to the laptop's manufacturer is another option. Additionally, YouTube can be a valuable resource, with numerous tech enthusiasts showcasing laptop internals, potentially revealing the presence of an SSD slot.

To ascertain if your laptop supports RAM upgrades without opening it, a simple method involves right-clicking the Start button, selecting Task Manager, navigating to the Performance tab, and then choosing the Memory section. This section displays the number of RAM slots and other pertinent memory information.

When aiming to purchase the ideal gaming laptop under the Rs 60,000 budget during forthcoming sales, it's imperative to combine thorough research with a comprehensive understanding of essential components. Elements such as GPU performance, VRAM, storage type, and RAM configuration are all crucial in guaranteeing an uninterrupted gaming experience. Furthermore, aspects like the efficiency of the cooling system and upgrade potential can greatly influence the laptop's durability and its ability to meet future gaming requirements. With the knowledge gleaned from this guide, you are now more prepared to make a well-informed choice and secure the most value during the sale. Here's to enjoyable gaming sessions ahead! Write to editor@skoar.in for any queries! *

